Practice Test C — Answer Key & Worked Solutions

One worked solution per question. Mark the working as well as the final answer.

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1. [Solving Linear Equations (for x)]

Answer: 6

Worked solution: Step 1: divide both sides by the coefficient 7. Step 2: x = 42 ÷ 7 = 6.

2. [Literal Equations & Solving for y]

Answer: A) $w = (P - 2l) \div 2$

Worked solution: Step 1: subtract 2l from both sides → $P - 2l = 2w$. Step 2: divide by 2 → $w = (P - 2l) \div 2$.

3. [Linear Inequalities]

Answer: A) $x \geq 5$

Worked solution: Step 1: add 6 to both sides → $3x \geq 15$. Step 2: divide by 3 (positive, no flip) → $x \geq 5$.

4. [Systems of Linear Equations]

Answer: A) (3, 2)

Worked solution: Elimination: Step 1: add the equations → 3x = 9. Step 2: x = 3. Step 3: plug back → 3 − y = 1 → y = 2. Solution: (3, 2).

5. [Functions: Notation, Domain & Range]

Answer: 6

Worked solution: Step 1: substitute 2 for x → h(2) = 5(2) − 4. Step 2: multiply → 10 − 4. Step 3: subtract → 6.

6. [Linear Functions: Slope, Intercepts & Equations]

Answer: 3

Worked solution: Step 1: slope = $(y_2 - y_1) \div (x_2 - x_1) = (8 - 2) \div (3 - 1)$. Step 2: = 6 ÷ 2 = 3.

7. [Quadratic Functions & Equations]

Answer: A) downward

Worked solution: Step 1: the leading coefficient $a = -1$ is negative. Step 2: a negative a opens the parabola downward.

8. [Polynomials: Operations & Factoring]

Answer: 6

Worked solution: Step 1: the coefficients 6 and 12 share a factor of 6. Step 2: the GCF is 6, so 6x + 12 = 6(x + 2).

9. [Radical & Rational Expressions]

Answer: A) least common denominator (LCD)

Worked solution: Multiplying every term by the LCD clears all the fractions so you can solve the resulting equation.

10. [Exponential Functions & Growth/Decay]

Answer: 25

Worked solution: Step 1: substitute 2 for x → $y = 100 \cdot (1/2)^2$. Step 2: $(1/2)^2 = 1/4$. Step 3: 100 × (1/4) = 25.

11. [Data: Measures of Center & Spread]

Answer: A) Mean

Worked solution: The mean uses every value in the sum, so a single extreme outlier pulls it strongly. The median and mode are more resistant.

12. [Scatter Plots, Line of Best Fit & Correlation]

Answer: A) strong negative correlation

Worked solution: r near −1 means a strong linear relationship where y decreases as x increases.

13. [Absolute Value Equations & Inequalities]

Answer: 4

Worked solution: Case 1: $2x + 1 = 9 \rightarrow 2x = 8 \rightarrow x = 4$. Case 2: $2x + 1 = -9 \rightarrow 2x = -10 \rightarrow x = -5$. The larger solution is 4. Check: $|2(4) + 1| = |9| = 9$ ✓.

14. [Arithmetic & Geometric Sequences]

Answer: B) 3

Worked solution: Divide a term by the one before it: $6 \div 2 = 3$, and $18 \div 6 = 3$, and $54 \div 18 = 3$. The common ratio is 3.

15. [Inverse Functions]

Answer: A) $\sqrt{x}$

Worked solution: Restricting the domain to $x \geq 0$ makes squaring one-to-one, so it has an inverse: the non-negative square root, $f^{-1}(x) = \sqrt{x}$. Check: $(\sqrt{x})^2 = x$ for $x \geq 0$ ✓.

16. [Function Transformations & Vertex Form]

Answer: A) reflects it across the x-axis

Worked solution: The minus sign outside the function negates every output: a point (x, y) becomes (x, −y). That is a reflection across the x-axis. f(−x) would be the reflection across the y-axis.

17. [Systems of Linear Inequalities & Linear Programming]

Answer: B) (5, 0)

Worked solution: Evaluate P at each vertex: (0, 0) → 0; (5, 0) → 15; (0, 4) → 8; (2, 3) → 6 + 6 = 12. The largest value is 15, at (5, 0).

18. [Probability from Two-Way Tables]

Answer: A) 0.15

Worked solution: “And” means the single cell where both are true: 30 students. Divide by the grand total, P = 30/200 = 0.15.

19. [Quadratic Functions & Equations]

Answer: 2

Worked solution: Step 1: a = 1 and b = −4. Step 2: $x = -(-4) \div (2 \cdot 1) = 4 \div 2 = 2$.

20. [Absolute Value Equations & Inequalities]

Answer: D) No solution

Worked solution: An absolute value is a distance, so it can never equal a negative number. There is no value of x that makes $|x - 4|$ equal −2, so the equation has no solution.

21. [Arithmetic & Geometric Sequences]

Answer: 160

Worked solution: $a_1 = 5$ and r = 2. $a_n = a_1 r^{n-1}$ gives $a_6 = 5 \cdot 2^5 = 5 \cdot 32 = 160$. Check by listing: 5, 10, 20, 40, 80, 160 ✓.

22. [Inverse Functions]

Answer: C) f(x) = 2x + 5

Worked solution: f(x) = 2x + 5 is a straight line with a non-zero slope, so it passes both the vertical and the horizontal line test. x^2 and |x| fail the horizontal line test (two inputs share an output), and a constant function sends every input to the same output, so neither can have a function inverse.

23. [Function Transformations & Vertex Form]

Answer: A) horizontal compression by a factor of 1/2

Worked solution: Inside changes act on x and run opposite to what you see. Multiplying the input by 2 makes the graph reach each value twice as fast, so it compresses horizontally by a factor of 1/2.

24. [Systems of Linear Inequalities & Linear Programming]

Answer: 30

Worked solution: The vertices are (0, 0), (10, 0) and (0, 10). P = 0 at (0, 0), P = 20 at (10, 0) and P = 3(10) = 30 at (0, 10). The largest value is 30.

Marking your test

- One mark per question, 24 marks in total. Give the mark only if the working supports the answer.
- 20 or more: the topics are solid — move on to timed mixed practice.
- 15 to 19: reread the method for each topic you dropped a mark in, then redo those questions in the app.
- 14 or fewer: work through the topic list in the app one topic at a time, then take the next practice test.
- Re-take any test after a week. If your score has not moved, that topic needs a different explanation, not more repetition.